Analysis of hILC RNA-seq data in patients of severe and mild dengue fever

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BCB420 A3 Presentation

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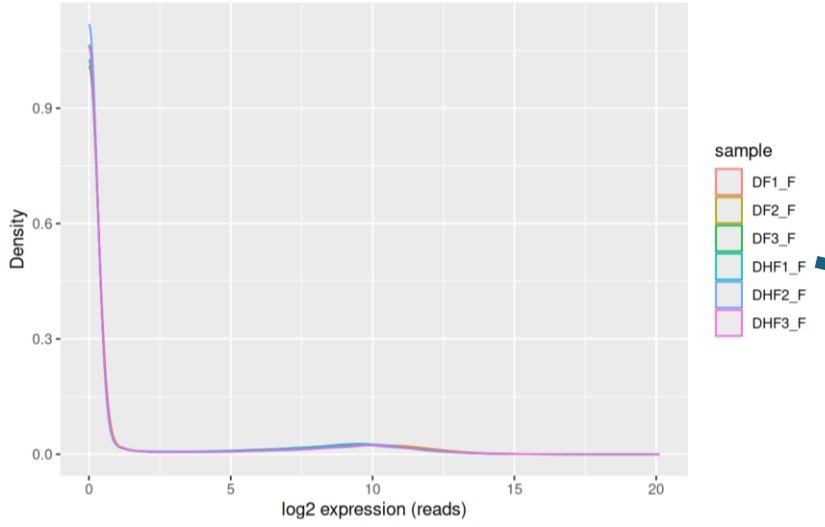
Introduction

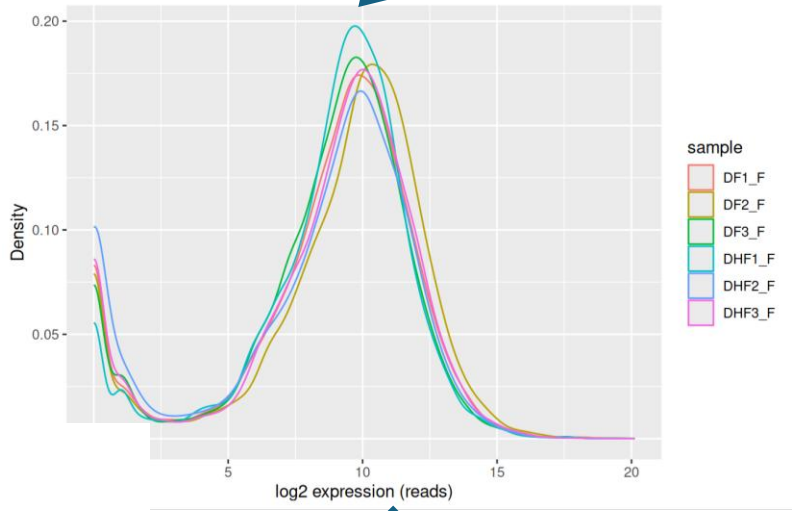
- Dengue virus remains a threat, especially tropical areas.
- Infections can range in severity, from mild dengue fever (DF) to dengue haemorrhagic fever (DHF), and dengue shock syndrome which can be fatal.
- Adaptive immune response has been characterized well, but studies on innate immune response are limited
  - This includes helper innate lymphoid cells (hILCs) which are akin to the helper T cells.

Introduction

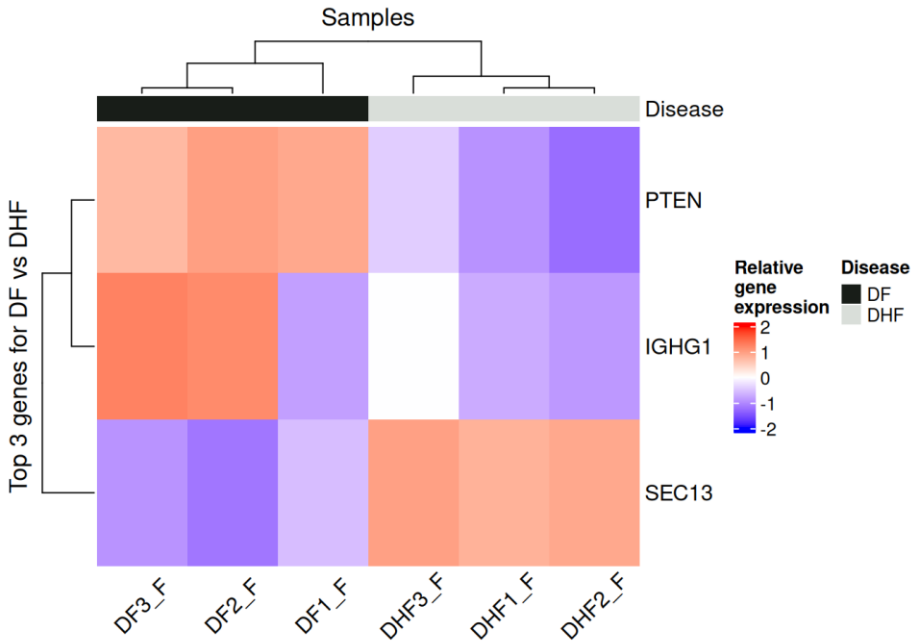
Bulk RNA-seq data of hILCs in patients in the **febrile** phase of the **severe dengue haemorrhagic fever (DHF),** or **mild dengue fever (DF)** (Poonpanichakul et al. (2021), GEO accession GSE155672).



Filtering low counts
TMM normalisation
(edgeR)



Differential expression Analysis (DHF vs DF)
- Scoring genes by logFC



Enrichment analysis method

Gene list (ranked by logFC)

Pathway gene sets (Bader Lab, March 02 2026 ver.)

Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

Home > Bioconductor 3.22 > Software Packages > **fgsea**

fgsea

This is the **released** version of fgsea; for the devel version, see fgsea.

Fast Gene Set Enrichment Analysis

fgsea_1.34.2

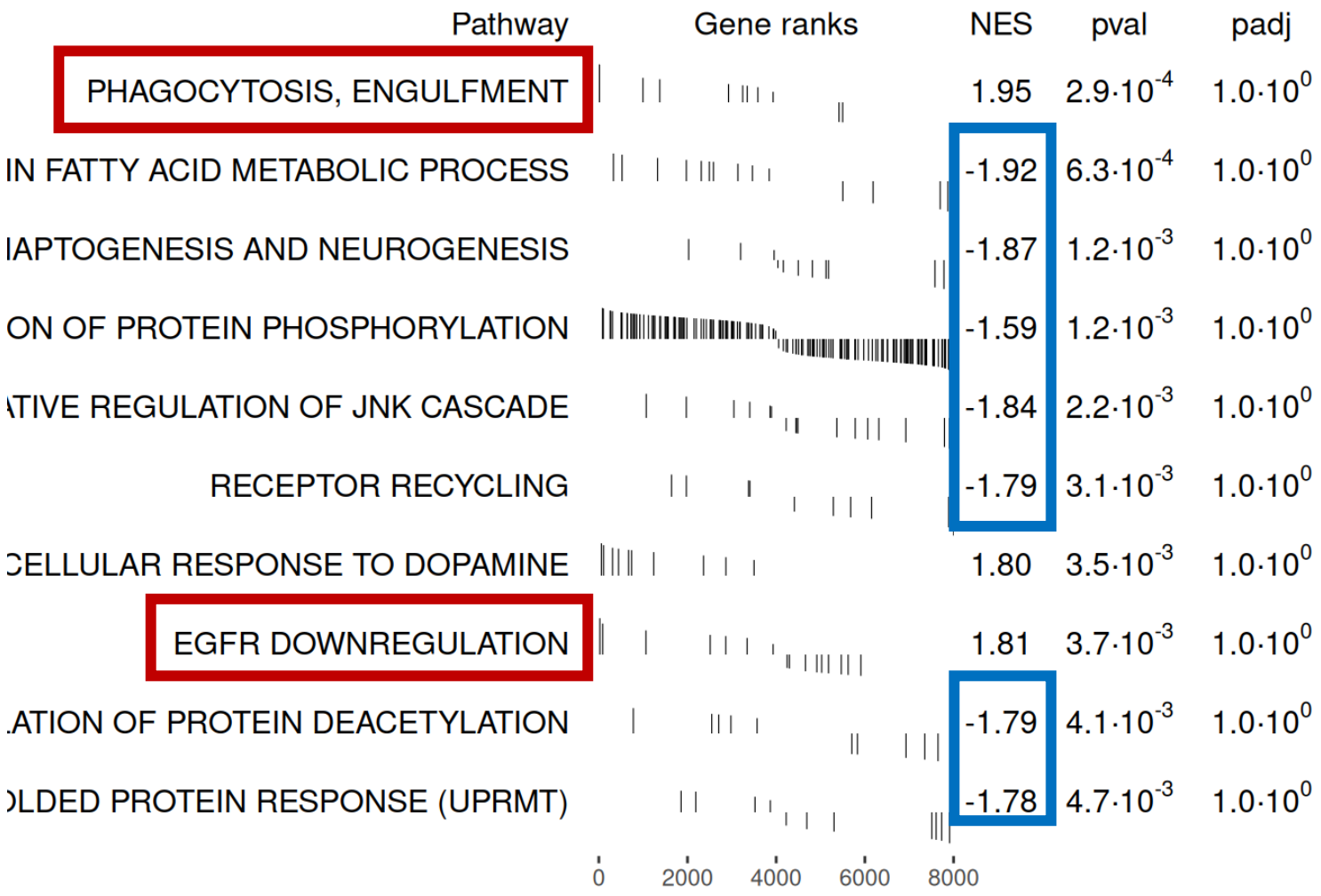
(Subramanian et al., 2005; Korotkevich et al., 2019)

Cytoscape version: 3.10.4
Enrichment Map version: v3.5.0
AutoAnnotate version: v1.5.2

(Shannon et al., 2003; Merico et al., 2010; Kucera et al., 2016)

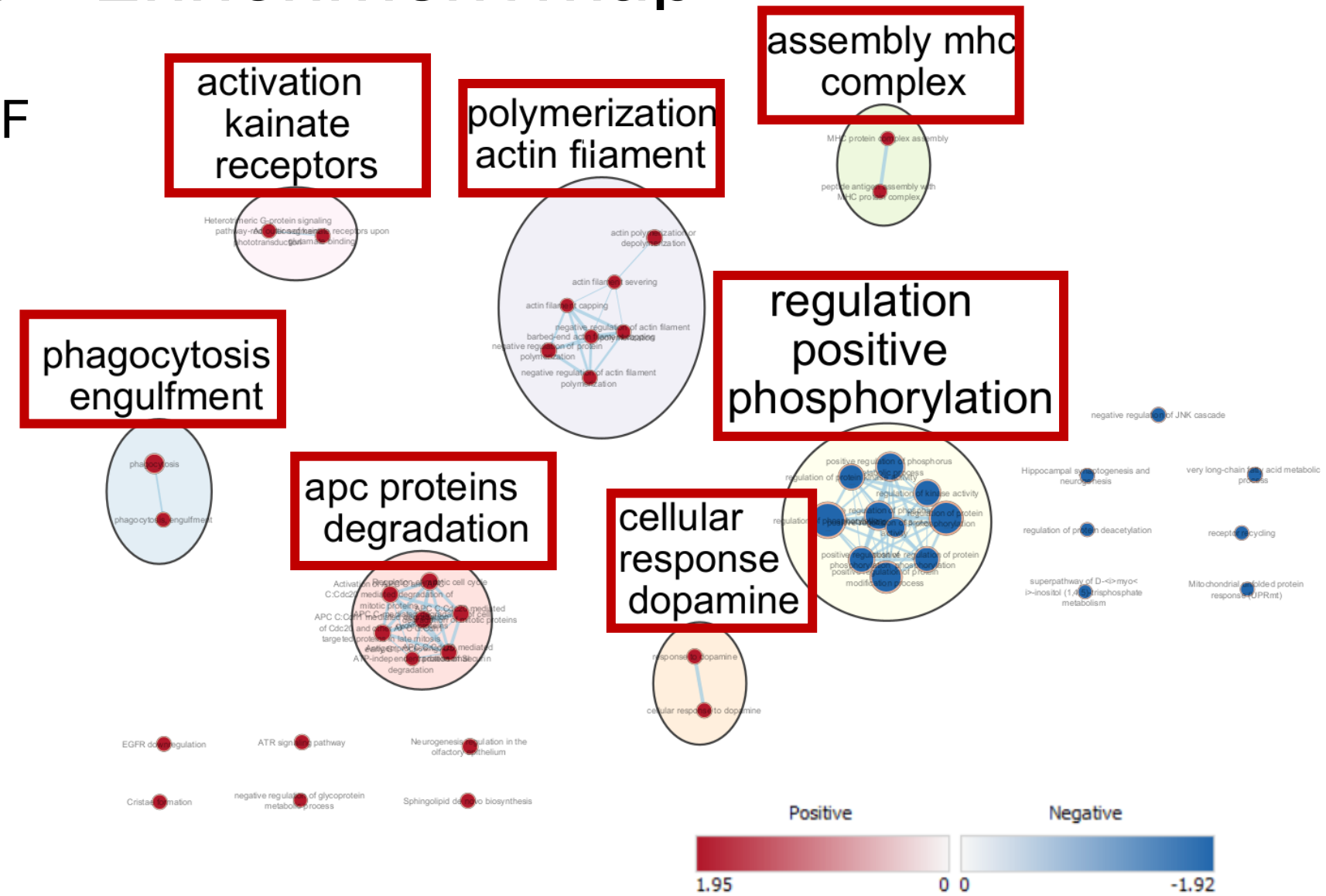
GSEA Results

- Phagocytosis, engulfment and EGFR downregulation in upreg. genes
- Among the top results (based on NES and p-value), most are enriched among downreg. genes



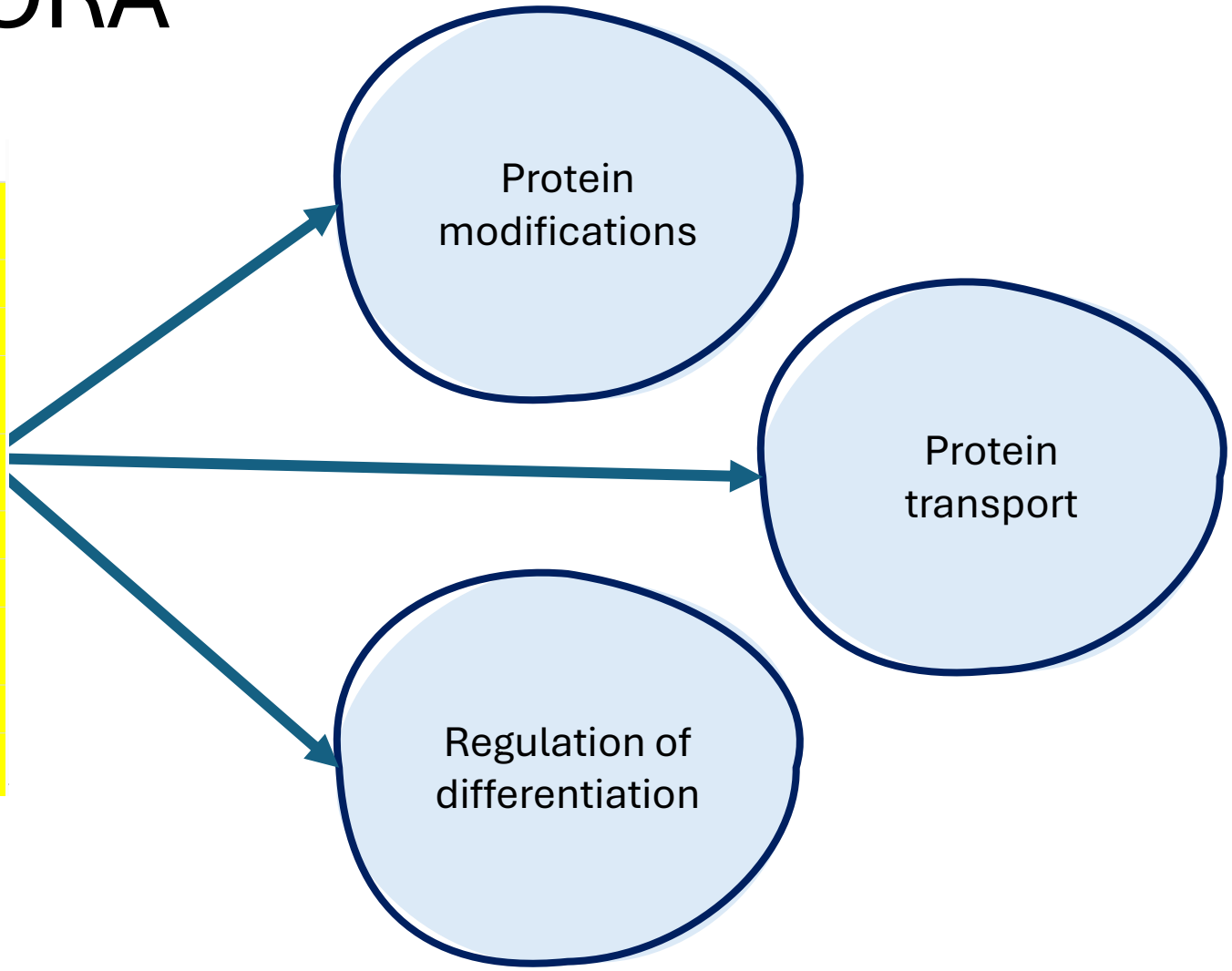
GSEA Results – Enrichment map

- Less regulation in DHF hILCs than in DF
- Macrophage activity
- Assembly of MHC complex
- Neurotransmitter-related



Comparison with ORA

p_value	term_size	intersection_size	source	term_name
0.0040880	12	4	GO:BP	positive regulation of protein acetylation
0.0173489	18	4	GO:BP	regulation of protein acetylation
0.0202133	151	10	GO:BP	peptidyl-serine phosphorylation
0.0286030	**21**	**4**	**GO:BP**	**DNA repair-dependent chromatin remodeling**
0.0302935	**80**	**7**	**GO:BP**	**positive regulation of myeloid cell differentiation**
0.0332300	**82**	**7**	**GO:BP**	**protein acylation**
0.0332300	163	10	GO:BP	peptidyl-serine modification
0.0472835	**174**	**10**	**GO:BP**	**nucleobase-containing compound transport**
0.0472835	89	7	GO:BP	acyl-CoA metabolic process
0.0472835	116	8	GO:BP	nuclear-transcribed mRNA catabolic process



Key insights – implications from results

- Correlation between lower regulation in hILC activation and DENV infection severity.
- Affirmation that hILCs play a role in different immune responses.
- Synaptic transmission-related genes may also be involved.
- However: high FDR – noisy data?
  - More samples, or could try different processing methods